INCH-POUND

MIL-T-27/209A
20 April 1993
SUPERSEDING
MIL-T-27/209
28 December 1977

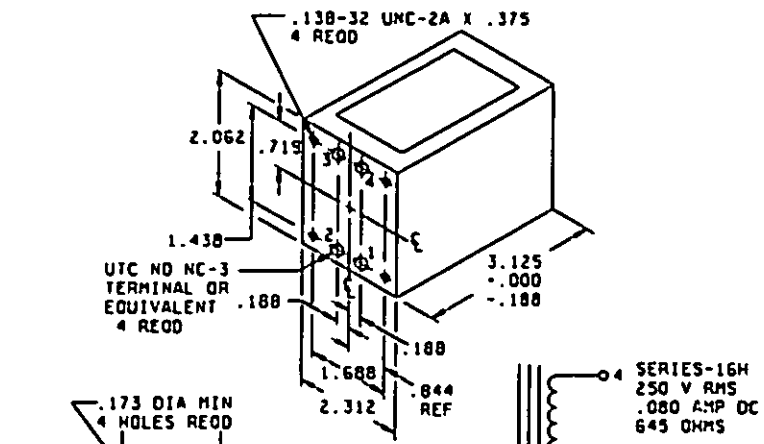
MILITARY SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS (AUDIO, POWER AND HIGH-POWER PULSE) INDUCTOR, POWER, TF4RD4FA

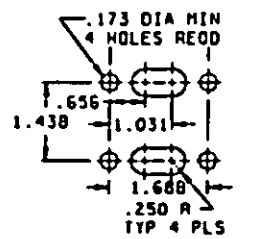
(A) Inactive for new design after the date of this specification.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.



Inches	mm
.178	4.39
.188	4.78
.250	6.35
.375	9.52
.656	16.66
.719	18.26
.844	21.44
1.031	26.19
1.438	36.53
1.688	42.88
2.062	52.37
2.312	58.72
3.125	79.38



MINIMUM CHASSIS CUTOUT

WORKING VOLTAGE (PEAK): 1000 V

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance on case dimensions is +.000 (0.00 mm), -.125 (3.18 mm).
4. Tolerance on terminal positioning dimensions is ±.125 (3.18 mm). Terminals fit within minimum chassis cutout.
5. Marking shall be on the sides of the case.

FIGURE 1. Dimensions and configurations.

(A) denotes changes

AMSC N/A

FSC 5950

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the windings.)

Electrical ratings:

Inductance (1-4):

Series connected: 16 H minimum.
Parallel connected: 4 H minimum.

Voltage (1-4):

Series connected: 250 volts rms.
Parallel connected: 125 volts rms.

Current (1-4):

Series connected: 0.080 ampere dc.
Parallel connected: 0.160 ampere dc.

DC resistance:

(1-2): 324.5 ohms, maximum.
(3-4): 385 ohms maximum.

Working voltage (peak): 1,000 volts.

Power: 2.08 watts maximum (each winding).

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal encased.

Material: Steel.

Terminals: Solder terminal, United Transformer Co. NC-3 or equivalent.

Terminal height: .469 inch, maximum.

Weight: 2 pounds, maximum.

Operating temperature range: -55°C to +105°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds and test condition E, 8 ounce-inches.

Dielectric withstanding voltage (at sea level): 2,400 volts rms.

Electrical characteristics:

Inductance:

Series connected: With 250 volts rms, 120 Hz, and 80 mA dc applied to (1-4), the inductance shall be 16 H, minimum.

Parallel connected: With 125 volts rms, 120 Hz, and 160 mA dc applied to (1-4), the inductance shall be 4 H, minimum.

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DC resistance:

(1-2): 324.5 ohms, maximum.
(3-4): 385 ohms, maximum.

Temperature rise: 40°C with 250 volts rms, 120 Hz, and 80 mA dc applied to (1-4) with (2-3) connected at an ambient temperature of 65°C.

Shock (specified pulse): MIL-STD-202, method 213, test condition H.

(A) Quality assurance provisions:

Qualification inspection: Not applicable for this specification sheet.

Quality conformance inspection: Groups A and B tests of MIL-T-27 shall be applicable.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/209-01.

NOTE: MIL-T-27/209 supersedes MS90009 which is inactive for new design after 28 December 1977. M27/209-01 supersedes MS90009-2.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 17, 99
DLA - ES

User activities:
Army - AR, ME, WC
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0815-19)

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